Choosing and Evaluating Models

Here we learn **"Build the model"** and **"Evaluate & critique model"** stages

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Mapping Problems to Machine Learning

- Our primary task is to connect a business problem to the right machine learning approach.
- This mapping is a critical first step.
- The type of business problem will determine whether we need to classify, score, or group data.
- For example, predicting if a customer will churn is different from predicting how much they will spend.

The Three Main ML Tasks

- **Classification**: Assigning known labels to objects.
- **Scoring (Regression)**: Predicting a numerical value.
- **Grouping (Unsupervised Learning)**: Finding patterns and structure in data.

Classification in Detail

- **Goal**: To predict a discrete category or class.
- **Example**: Identifying if a financial transaction is fraudulent (Yes/No).
- **Supervised Learning**: This is a type of supervised learning, meaning we need a training dataset with known labels to build the model.
- The model learns to map input features to a specific output label.

Types of Classification

- **Two-Category (Binomial) Classification**: The simplest form, with only two possible outcomes.
    - *Examples*: Spam vs. Not Spam, Customer Churn vs. No Churn.
- **Multicategory Classification**: More than two possible outcomes.
    - *Examples*: Classifying a handwritten digit as 0-9, Categorizing a product as "Electronics," "Apparel," or "Home Goods."

Scoring (Regression) in Detail

- **Goal**: To predict a continuous numerical value.
- **Example**: Predicting the price of a house based on its size, location, and number of bedrooms.
- **Supervised Learning**: Like classification, this is also supervised. We need a training set with known numerical outcomes.
- The model learns the relationship between input variables and the numerical output.

Prediction vs. Forecasting

- While often used interchangeably, there is a subtle difference.
- **Prediction**: Choosing a specific, discrete outcome. For example, predicting if an email is spam.
- **Forecasting**: Assigning a probability to an outcome. For example, forecasting the probability of rain tomorrow.
- Scoring models can be used for both.

Grouping (Unsupervised Learning)

- **Goal**: To discover inherent structure or patterns in data without pre-existing labels.
- **Unsupervised Learning**: The model is not given a "right answer" to learn from.
- This is useful for exploratory data analysis and finding hidden relationships.

Types of Grouping

- **Clustering**: Grouping similar data points together.
    - *Example*: Segmenting a company's customers into different groups based on their purchasing behavior to tailor marketing strategies.
- **Association Rules**: Finding items that tend to occur together.
    - *Example*: In a supermarket, discovering that customers who buy diapers also tend to buy baby wipes.

The Importance of Model Evaluation

- We need to know how good our model is before we deploy it.
- Evaluation helps us choose the best model among several candidates.
- It also helps us identify problems like **overfitting**.
- Proper evaluation gives us confidence that the model will work on new, unseen data.

The Null Model

- Before building a complex model, we should define a **null model**.
- A null model is a simple, constant model that represents a baseline performance.
    - *For a classification problem*: The null model might always predict the most common class.
    - *For a scoring problem*: The null model might always predict the average value of the outcome.
- Our new model must perform better than the null model to be considered valuable.

Overfitting: A Major Pitfall

- **Overfitting** occurs when a model learns the training data too well, including its noise and quirks.
- The model will have excellent performance on the training data, but poor performance on new data.
- We can spot this when the **training error** is much lower than the **generalization error**.

Data Partitioning

- To avoid overfitting and get an unbiased evaluation, we need to split our data.
- The most common method is the **training/test split**.
- The training set is used to build the model.
- The test set is used *only* to evaluate the final model's performance on unseen data.
- A good split is often 70% training and 30% test.

K-Fold Cross-Validation

- When our dataset is small, a simple training/test split may not be enough.
- **K-fold cross-validation** is a more robust method.
- The data is divided into **k** equal-sized folds.
- The model is trained **k** times, each time using k-1 folds for training and one fold for testing.

The Confusion Matrix

- The **confusion matrix** is a fundamental tool for evaluating classification models.
- It's a table that summarizes the performance of a classifier by comparing predicted and actual labels.
- It helps us visualize which classes are being predicted correctly and which are being "confused" with others.

- Given a sample of 12 individuals, 8 that have been diagnosed with cancer and 4 that are cancer-free, where individuals with cancer belong to class 1 (positive) and non-cancer individuals belong to class 0 (negative), we can display that data as follows:

Individual number	1	2	3	4	5	6	7	8	9	10	11	12
Actual classification	1	1	1	1	1	1	1	1	0	0	0	0
Predicted classification	0	0	1	1	1	1	1	1	1	0	0	0

Individual number	1	2	3	4	5	6	7	8	9	10	11	12
Actual classification	1	1	1	1	1	1	1	1	0	0	0	0
Predicted classification	0	0	1	1	1	1	1	1	1	0	0	0
Result	FN	FN	TP	TP	TP	TP	TP	TP	FP	TN	TN	TN

		Predicted condition	
	Total population = P + N	Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

<table>
<tr><td></td><td></td><td colspan="2">Predicted condition</td></tr>
<tr><td></td><td>Total
8 + 4 = 12</td><td>Cancer (7)</td><td>Non-cancer (5)</td></tr>
<tr><td rowspan="2">Actual condition</td><td>Cancer
8</td><td>6</td><td>2</td></tr>
<tr><td>Non-cancer
4</td><td>1</td><td>3</td></tr>
</table>

- **Precision:**

  Calculates the proportion of true positive predictions out of all instances predicted as positive (True Positives / (True Positives + False Positives))

  Recall:

  Calculates the proportion of true positive predictions out of all actual positive instances (True Positives / (True Positives + False Negatives)).

Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

https://en.wikipedia.org/wiki/Confusion_matrix

	Actual 0	Actual 1
Predicted 0	True Negatives (TN)	False Negatives (FN)
Predicted 1	False Positives (FP)	True Positives (TP)

It looks something like this (considering 1 -Positive and 0 -Negative are the target classes):

- TN: Number of negative cases correctly classified
- TP: Number of positive cases correctly classified
- FN: Number of positive cases incorrectly classified as negative
- FP: Number of negative cases correctly classified as positive

- ACCURACY
- Accuracy answers the question, “When the spam filter says this email is or is not spam, what’s the probability that it’s correct?”

precision

- Precision answers the question, “If the spam filter says this email is spam, what’s the probability that it’s really spam?” Precision is defined as the ratio of true positives to predicted positives.
- Precision: How often a positive indication turns out to be correct

--- How often a positive indication turns out to be correct.

- Recall: What fraction of the things that are in the class are detected by the classifier

Key Classification Metrics

- **Precision**: `(True Positives) / (True Positives + False Positives)`
  - Measures how many of the positive predictions were actually correct.
- **Recall**: `(True Positives) / (True Positives + False Negatives)`
  - Measures how many of the actual positive cases were correctly identified.
- **F-score**: A metric that combines precision and recall. It's the harmonic mean of the two. A high F-score indicates both high precision and high recall.

True positive rate (TPR), recall, sensitivity (SEN), probability of detection, hit rate $= TP/P$

True negative rate (TNR), specificity (SPC), selectivity $= TN/N = 1 - FPR$

FPR=False positive rate

Evaluating Scoring Models

- For regression (scoring) models, we use different metrics.
- **Mean Squared Error (MSE)**: Calculates the average of the squared differences between the predicted and actual values.
- **Root Mean Squared Error (RMSE)**: The square root of the MSE. It is in the same units as the outcome variable, making it easier to interpret.
- A lower RMSE or MSE indicates a more accurate model.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{n}(Y_i - \overline{Y_i})^2}$$

- **y_i** is the i^{th} observation, and
- yi^ is the estimated value of y_i.
- $R^2 = 1 - (RSS/TSS)$

- The best possible R-squared is 1.0, with near-zero or negative R-squareds being horrible.
- https://mlu-explain.github.io/linear-regression/

Evaluating Probability Models

- For models that output probabilities, we use metrics like **AUC** (Area Under the Curve).
- **AUC** measures the model's ability to distinguish between the positive and negative classes.
- An AUC of 1.0 means the model can perfectly distinguish between the two classes. An AUC of 0.5 means the model is no better than random guessing.

LIME: Explaining Your Model

- **LIME (Local Interpretable Model-agnostic Explanations)** is a package that helps us understand *why* a model made a specific prediction.
- It provides a "local" explanation for a single prediction, not a global one for the whole model.
- This is a crucial tool for **model debugging and auditing**.
- If a model makes a questionable prediction, LIME can show us which features led to that decision, helping us uncover biases or flaws in our data.

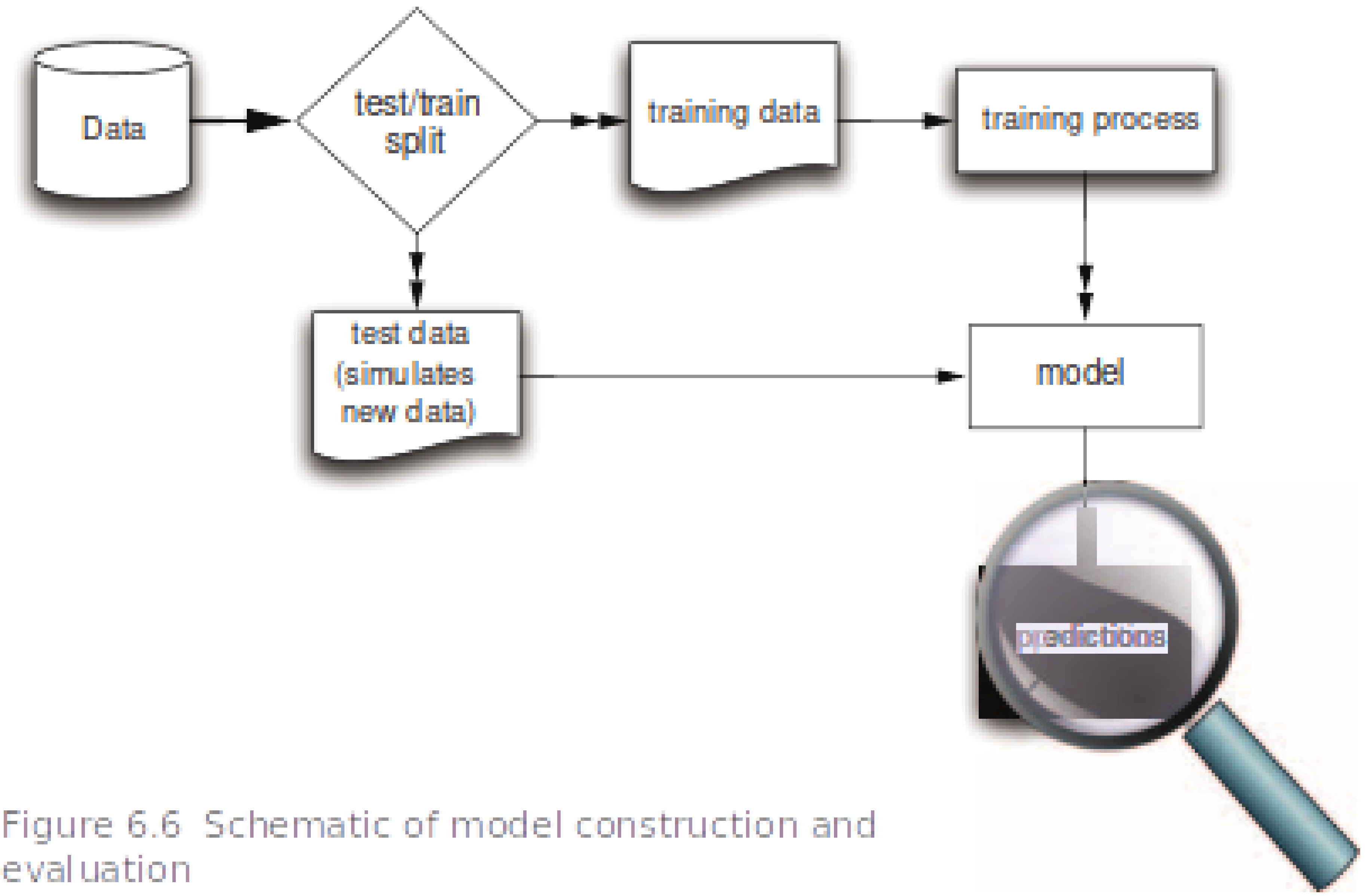
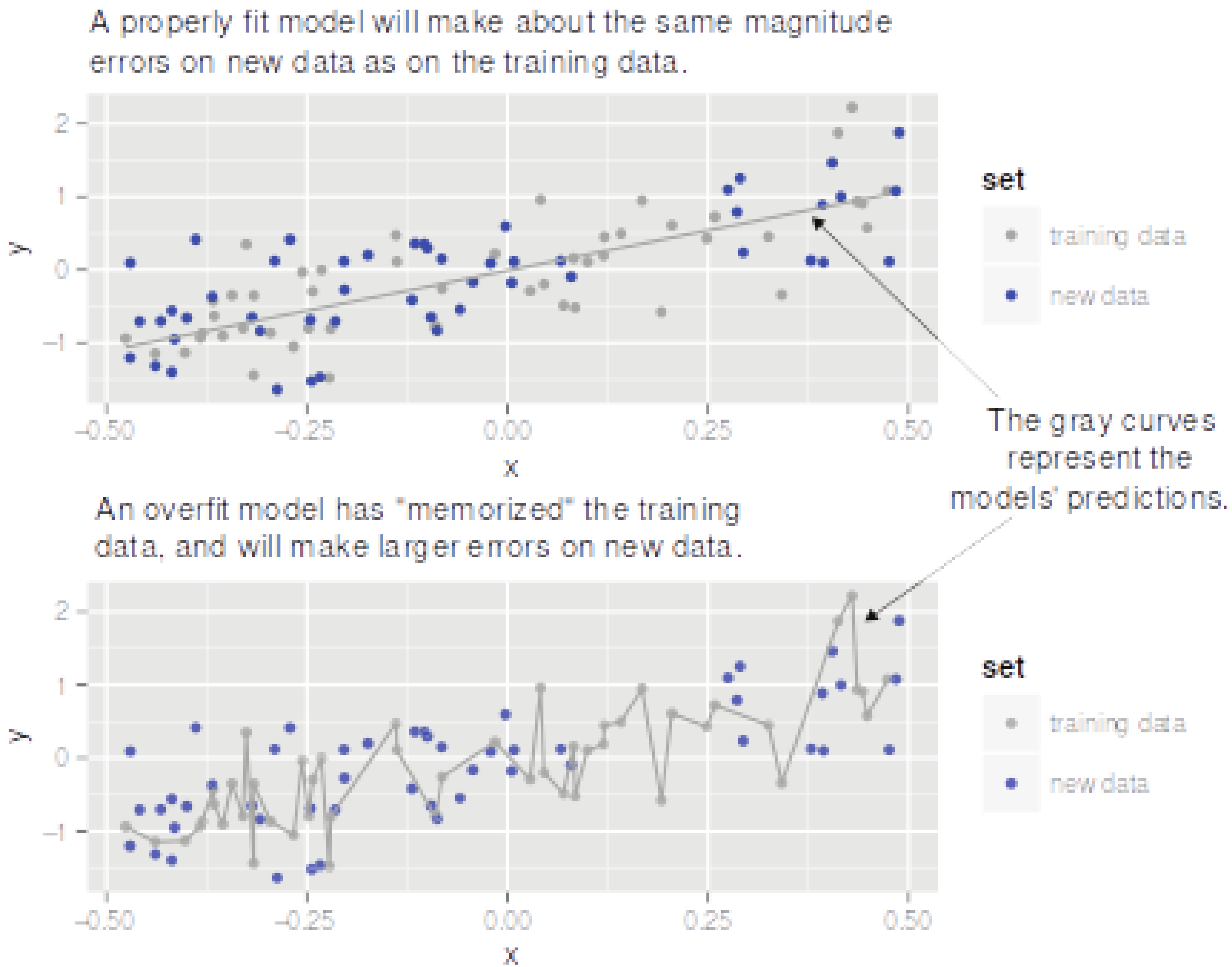


Figure 6.6 Schematic of model construction and evaluation


A properly fit model will make about the same magnitude errors on new data as on the training data.
An overfit model has "memorized" the training data, and will make larger errors on new data.
The gray curves represent the models' predictions.
y
x
2
1
0
−1
−0.50
−0.25
0.00
0.25
0.50
set
training data
new data

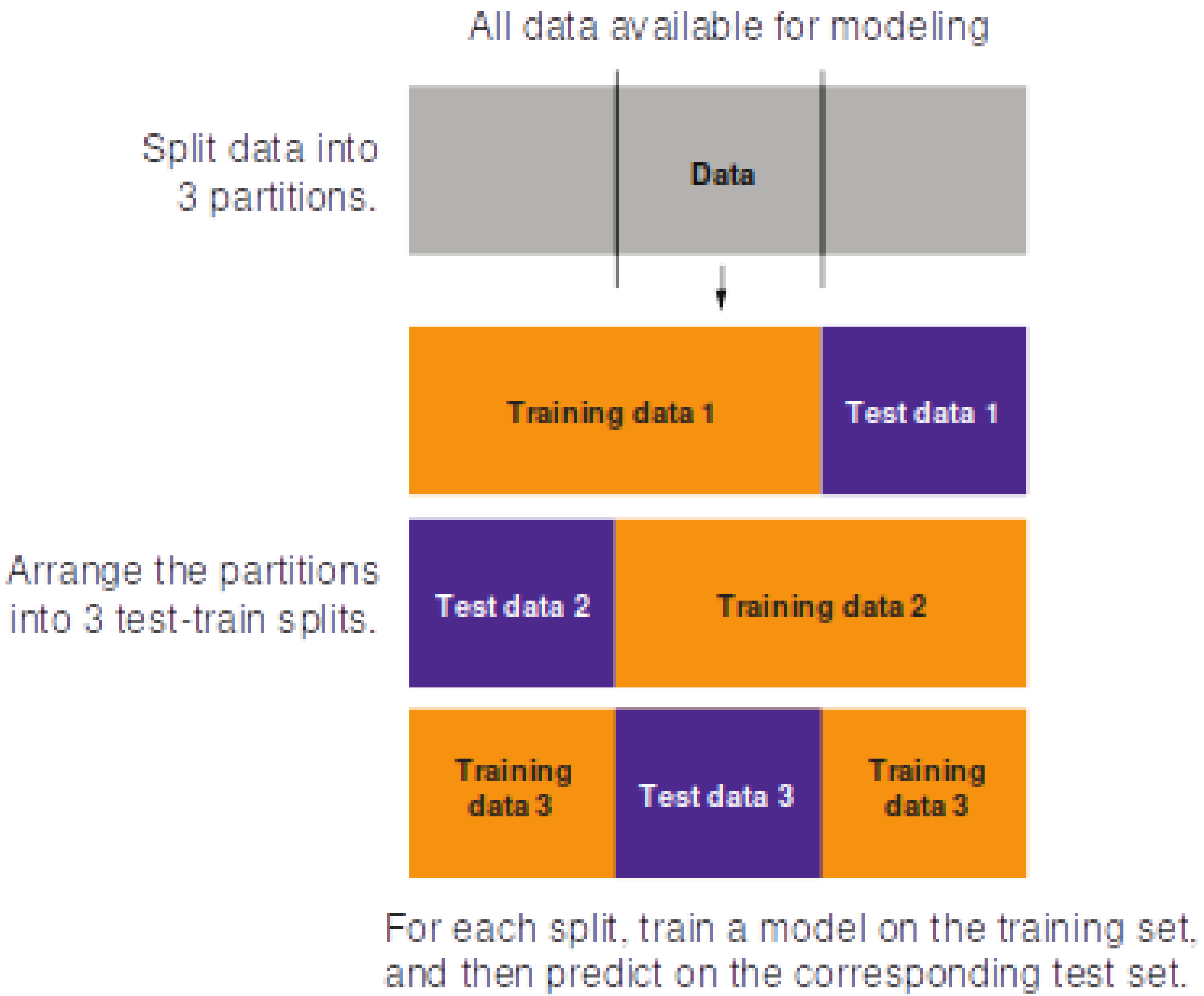


Figure 6.9 Partitioning data for 3-fold cross-validation